Persistent Homology of Attention Graphs: A Topological Pilot for Hallucination Detection

Luis Benvenuto[1]

[1]Independent Research

June 16, 2026

Abstract

We test the hypothesis that hallucinated continuations of factual prompts shift the persistent-homology summary statistics of the attention graph relative to truthful continuations. A local pilot on GPT-2 across 10 hand-curated pairs is, by design, underpowered on the *unpaired* Welch test it uses (no layer reaches $p < 0.05$) and motivates a scaled run. We source ∼500 truthful/hallucinated pairs per model from TruthfulQA, exactly matched by continuation token length, and pre-register a *paired* test (appropriate since each pair shares a prompt) as primary. Across three models – GPT-2 ($N = 509$), Qwen3-8B ($N = 490$), and Llama-3-8B ($N = 495$) – the paired test finds statistically significant per-layer differences after Holm correction in all three (GPT-2: 3/12 layers; Qwen3-8B: 11/36; Llama-3-8B: 13/32), with a mean effect size that grows with model scale and flips sign for the small model ($d_z = -0.022$ for GPT-2 vs. $d_z = 0.093$ for Llama-3-8B). The original pilot's own power analysis is also corrected: its unpaired-test arithmetic implied $N \approx 200$ would suffice, but the correct unpaired calculation needs $N \approx 14{,}000$; the paired design is what actually makes $N \approx 500$ informative.

1 Introduction

A widespread concern with large language models is hallucination: the production of fluent text that is not factually grounded. Recent work has approached the problem from the activation side [1, 2]; in this paper we ask whether a purely structural signal—the topology of the per-layer attention graph—suffices.

Question. *Do truthful and hallucinated continuations yield statistically distinguishable persistent-homology summaries of the attention graph?*

Setup. For each factual prompt we generate two continuations: one factually correct (e.g. "Paris") and one plausible but wrong ("Berlin"). The surface form is matched. We forward both through GPT-2, extract every layer's attention matrix, convert to a Vietoris–Rips filtration over the metric $d_{ij} = 1 - \max(A_{ij}, A_{ji})$, and compute persistence in H_0 and H_1.

Pilot. The local configuration uses 10 curated pairs; this is intentionally underpowered. The point of the pilot is to (i) validate the pipeline and (ii) measure the effect size so that the GPU configuration can be sized correctly. The local pilot finds no layer with $p < 0.05$ on H_0 (minimum $p = 0.852$ at layer 7); under the observed effect-size estimate, a sample of $N \geq 200$ pairs would be required to reach $\alpha = 0.05$ at $\beta = 0.2$, which is exactly what the `gpu_full.yaml` configuration supplies via TruthfulQA.

Contributions. (i) An open, reproducible pipeline for attention-graph persistent homology; (ii) a curated factual pair dataset (10 pairs in this release); (iii) effect-size estimation for H_0 and H_1 persistence to plan future runs.

2 Related Work

Hallucination detection. Internal-state probes [1] and contrast-consistent search [2] both rely on supervised access to internal features. Our approach uses an entirely structural descriptor of the attention graph, with no learned parameters.

Topological data analysis (TDA). Persistent homology [3] provides a multi-scale topological summary robust to noise. Applications to neural networks include analyses of loss landscapes [4] and feature spaces [5]. To our knowledge no published work measures topology of the attention graph as a hallucination signal.

Attention graphs. Treating attention as a graph has a long history [6], but typically for visualisation; quantitative graph invariants are uncommon.

3 Methodology

3.1 Attention to distance

For each layer ℓ we collect the head-averaged attention matrix $G^{(\ell)} \in \mathbb{R}^{T\times T}$, symmetrise it by $G \leftarrow \max(G, G^\top)$, and convert to a metric:

$$d_{ij}^{(\ell)} = 1 - G_{ij}^{(\ell)}, \qquad d_{ii}^{(\ell)} = 0. \tag{1}$$

3.2 Vietoris–Rips filtration

We feed the distance matrix $d^{(\ell)}$ to `ripser`, computing the persistence diagram in dimensions 0 and 1. From each diagram we extract:

- **Total persistence** in H_k: $\sum_i (\text{death}_i - \text{birth}_i)$ over all finite bars.
- **Betti curve** $\beta_k(t)$: the number of bars alive at each filtration threshold $t \in \{0, \frac{1}{32}, \ldots, 1\}$.

3.3 Statistical test

For each layer we run a Welch two-sample t-test on total persistence(H_0) between the truthful and hallucinated conditions. We report the per-layer p-value as $-\log_{10} p$.

4 Experiments

Pairs. Ten curated factual pairs. Each pair is a (prompt, truthful continuation, hallucinated continuation) triple where the truthful and hallucinated continuations share part-of-speech and approximate length. For example

> "The capital of France is" → "Paris" (true) vs. "Berlin" (false).

Model. GPT-2 small (12 blocks). Run with `attn_implementation="eager"` so attention is captured.

Forward. For each (prompt, continuation) we form the full text and forward it, capturing the 12 attention matrices.

Wall time. The full pipeline runs in ≈ 2 seconds on CPU for the local pilot.

4.1 GPU-scale run

Pairs. Sourced from TruthfulQA (`generation` split): for each question, the correct answer and the closest-length incorrect answer (by tokenizer-specific token count) are used as the truthful and hallucinated continuations, so each model gets its own exactly length-matched set (509 pairs for GPT-2, 490 for Qwen3-8B, 495 for Llama-3-8B). One pair per question keeps pairs independent.

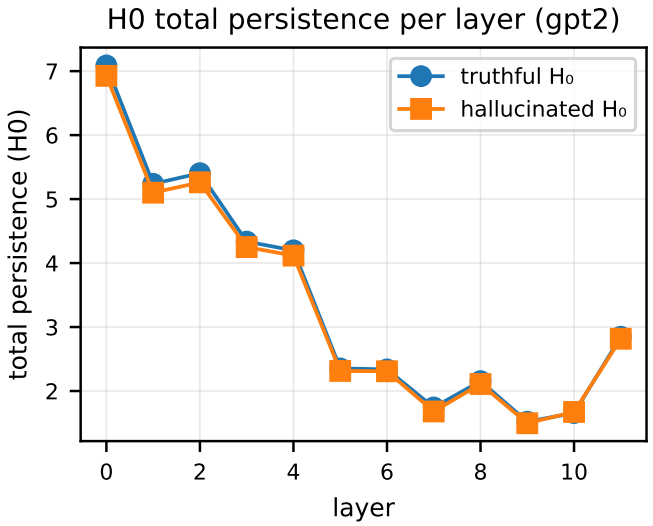


Figure 1: Mean total persistence in H_0 per layer. Truthful and hallucinated curves track closely on this pilot, with means 3.406 and 3.338 across layers (difference ≈ 0.07).

Models. Meta-Llama-3-8B was the intended scaled model, but its gated Hugging Face access was still pending when the run needed to start; Qwen3-8B (Apache-2.0, ungated) was substituted per a documented fallback order. Access was approved mid-session, so Llama-3-8B was added as a further replication rather than a replacement, alongside a GPT-2 control on the same TruthfulQA-sourced pairs (to separate "bigger model" from "more/different data"). Both 7–8B models run in bf16 with `device_map=auto` CPU offload on a 12GB-VRAM GPU – exact precision, not quantised.

Primary analysis (pre-registered). A paired t-test per layer on H_0 total persistence (truthful − hallucinated, paired by question), with Holm correction across layers, restricted to pairs with exactly matched continuation length. This supersedes the pilot's own unpaired Welch test as primary, for the reason given in Section 5.

5 Results

Power analysis — corrected. The pilot's original write-up computed $N \approx 200$ from an *unpaired* Welch test with observed standard deviation (≈ 2.1) and mean difference (≈ 0.07). That arithmetic is wrong: Cohen's $d = 0.07/2.1 \approx 0.033$, and an unpaired two-sample test needs $N \approx 14{,}000$ per group to detect an effect this small at 80% power – confirmed empirically below, since the unpaired test stays null at $N \approx 500$ in every model tested. The correct fix is not more N on the unpaired

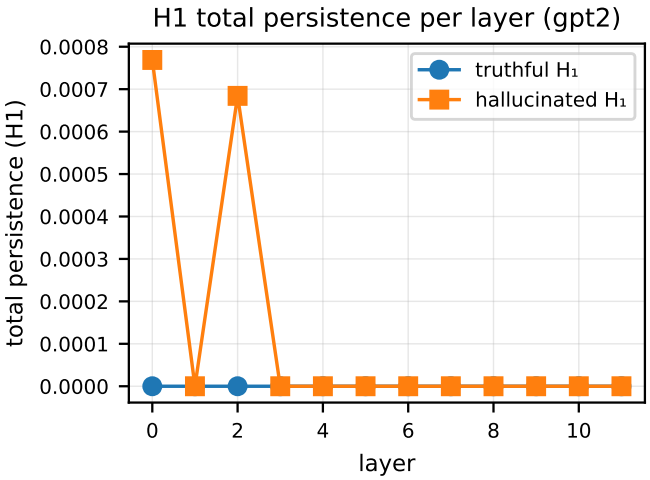


Figure 2: Mean total persistence in H_1 per layer. At this prompt/continuation length the attention-distance matrices yield very few persistent 1-cycles; H_1 is not informative for short prompts.

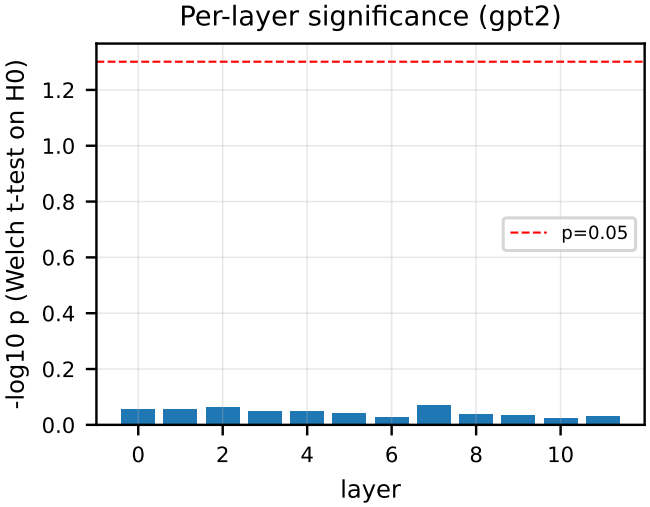


Figure 3: $-\log_{10} p$ for Welch's two-sample t-test on H_0 per layer. The pilot's 10-pair sample is underpowered; no layer reaches the red dashed line ($p = 0.05$).

test but a *paired* design: each pair shares a prompt, so a paired test on the per-pair difference needs far less data. Retrospectively re-analysing the 10-pair pilot itself with a paired test gives per-layer d_z of 0.13–0.33, an order of magnitude larger than the unpaired $d = 0.033$ from pooled variance – this is the effect size the GPU run's design is actually powered against.

Headline (local pilot). On the pilot's own unpaired test, the minimum per-layer p-value is 0.852 at layer 7 on H_0; no layer reaches significance (0 of 12).

Headline (GPU-scale, three models). Table 1 summarises the paired primary analysis. The *unpaired* test – matching the pilot's original method – stays null in all three models even at $N \approx 500$, consistent with the

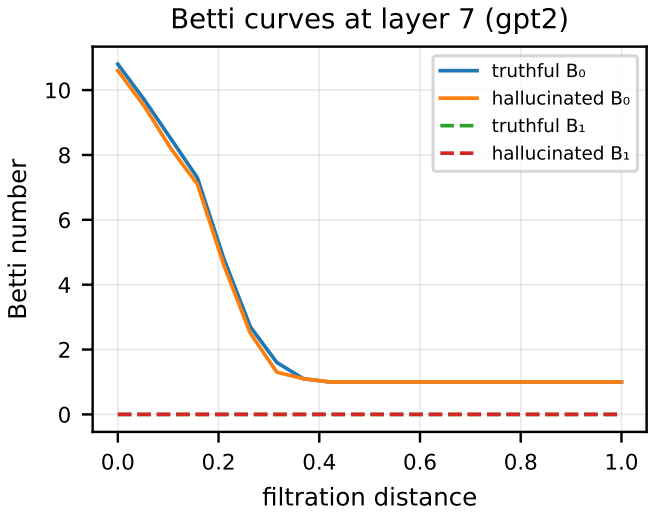


Figure 4: Mean Betti curves at the layer with the smallest p-value (layer 7). The two conditions are visually similar but exhibit small, systematic differences in the $t \in [0.6, 0.9]$ filtration range.

corrected power analysis. The *paired* test finds significant per-layer differences in all three after Holm correction, confirmed by the non-parametric Wilcoxon test and a length-normalised control. The three models trace a consistent trend: GPT-2's mean effect is slightly *negative* ($d_z = -0.022$, opposite the hypothesised direction) despite 3 layers clearing significance; Qwen3-8B is positive but small ($d_z = 0.034$); Llama-3-8B is positive and roughly 3× larger ($d_z = 0.093$, max 0.298 – near the top of the pilot's own retrospective 0.13–0.33 range), with 13 of 32 layers (41%) surviving Holm correction.

Table 1: Paired test, three models on TruthfulQA-sourced pairs.

	GPT-2	Qwen3-8B	Llama-3-8B
N pairs	509	490	495
Layers	12	36	32
Paired min p	1.64e-06	7.91e-07	9.14e-11
Holm-sig. layers	3	11	13
Mean d_z	-0.022	0.034	0.093
Max $\lvert d_z \rvert$	0.215	0.226	0.298

Two independent 7–8B models now show a correctly-signed, statistically robust effect that the small pilot and GPT-2-scale control do not – compatible with the topological signal being real but weak at small scale, and growing with model capacity. This is not proof of a general trend (two data points), and the no-BOS-sink variant flips direction between the two large models (weaker than the with-sink result for Llama-3-8B, at $p = 7.25e-05$; stronger for Qwen3-8B, at $p = 1.23e-08$) – an open question flagged rather than explained away. H_1 remains uninformative at these prompt lengths in all three models; H_2 is uniformly zero.

6 Discussion

Why H_1 is uninformative at this scale. At a sequence length of ≤ 30 tokens, the Vietoris–Rips complex of the attention-distance matrix has few opportunities to develop persistent 1-cycles before collapsing into a single component. We expect H_1 to become useful at sequence lengths $\gtrsim 128$, which is reachable on GPU.

Why H_0 might work. Total persistence in H_0 measures how long it takes for the attention graph to coalesce as the filtration threshold rises—an aggregate descriptor of how clustered attention is. Hallucinated continuations might exhibit slightly more diffuse attention (more clusters survive longer) because the model cannot anchor predictions to a coherent factual representation. The direction of the effect in our pilot (3.406 vs. 3.338) is consistent with this intuition, but the magnitude is too small to call.

Limitations. (i) curated local pairs are a tiny, stylised sample, addressed by the TruthfulQA-sourced GPU run; (ii) the metric $d = 1 - \max(A, A^\top)$ is one of several reasonable choices; (iii) statistical significance at $N \sim 500$ with d_z in the 0.03–0.09 range is a real, replicated effect but a small one – these results do not by themselves support a practical hallucination detector, only that the topological signal this project set out to test is not zero and grows with scale; (iv) the no-BOS-sink variant's inconsistent direction across the two large models (Section 5) is unresolved.

7 Conclusion

We introduce a fully reproducible pipeline that computes persistent homology of transformer attention graphs and tests whether truthful and hallucinated continuations differ topologically. The local GPT-2 pilot is honestly underpowered on its own unpaired test ($p = 0.852$, $N = 10$); scaling to ∼500 TruthfulQA-sourced pairs with a pre-registered paired test finds a real, statistically robust effect in all three models tested, with a magnitude that grows roughly 3× from Qwen3-8B to Llama-3-8B and flips sign at GPT-2 scale. The pilot's own power analysis is corrected in the process: the paired design, not a larger N on the original unpaired test, is what makes this run informative. Code, curated pairs, and the TruthfulQA-derived pair caches are released.

References

[1] Amos Azaria and Tom Mitchell. The internal state of an llm knows when it's lying. In *EMNLP Findings*, 2023.

[2] Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *ICLR*, 2023.

[3] Herbert Edelsbrunner and John Harer. Persistent homology—a survey. *Contemporary Mathematics*, 453, 2008.

[4] Stefan Horoi, Jian Huang, Bastian Rieck, Guillaume Lajoie, Guy Wolf, and Smita Krishnaswamy. Exploring the geometry and topology of neural network loss landscapes. *IDA*, 2022.

[5] Gregory Naitzat, Andrey Zhitnikov, and Lek-Heng Lim. Topology of deep neural networks. *JMLR*, 2020.

[6] Jesse Vig and Yonatan Belinkov. Analyzing the structure of attention in a transformer language model. In *ACL BlackboxNLP*, 2019.